

Deploying AI

Agents

```
$ echo "Data Sciences Institute"
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Main Points

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1. Agents extend foundation models into action-capable systems by combining reasoning, planning, and tool use in feedback loops.
2. The environment and available actions define what an agent can and cannot do; tool selection shapes both capability and risk profile.
3. Tools fall into three categories: knowledge augmentation (retrieval, search), capability extension (calculators, code interpreters), and write actions (emails, SQL updates, transactions).
4. Write actions are qualitatively riskier than read actions — agents that can modify their environment require explicit safeguards and human-in-the-loop oversight.

Main Points (cont.)

5. Planning is what separates agents from chatbots: good planners break complex tasks into subtasks, reason step by step, and recover from intermediate errors.
6. Errors compound across multi-step pipelines — a small per-step failure rate produces a large end-to-end failure rate, making agent reliability a distinct engineering challenge.
7. Agent evaluation must account for success rates across complete tasks, accumulated cost and latency, and safety outcomes for write actions — single-turn metrics are insufficient.

Agenda

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- Planning
- Interacting with APIs and MCP
- Agent failure modes and evaluation

Agents

Introduction to Agents

- An agent perceives its environment and acts upon it.
- Agents are defined by their environment and available actions.
- They can leverage tools to expand their capabilities.
- Chatbots with retrieval or browsing abilities are examples of agents.
- Agents combine reasoning, planning, and tool use.

Why Agents Matter

- Agents can automate workflows that require multiple steps.
- They integrate perception, planning, and action into cohesive loops.
- Examples include research assistants, trip planners, and negotiation bots.
- Agents represent a path toward autonomous, goal-driven AI systems.

Agent Components

- **Environment:** defines the context in which the agent operates.
- **Actions:** possible operations available to the agent.
- **Tools:** extend the agent's ability to perceive or act.
- **Planner:** determines how to sequence actions to reach goals.
- **Feedback loop:** updates decisions based on environment responses.

Tools for Agents

- Tools enhance agents by providing knowledge and capabilities.
- **Knowledge augmentation** tools retrieve or access data sources.
- **Capability extension** tools solve inherent model weaknesses, such as calculators.
- **Write actions** tools allow agents to alter environments, such as sending emails.
- Tool selection shapes the effectiveness and reliability of agents.

Knowledge Augmentation Tools

- Examples include text retrievers, image retrievers, and SQL executors.
- Tools can provide access to organizational databases and APIs.
- Web browsing tools help agents access up-to-date public information.
- APIs for search, news, and social media extend knowledge coverage.

Capability Extension Tools

- Address inherent model limitations, such as poor arithmetic skills.
- Simple extensions include calculators, calendars, and unit converters.
- More advanced tools include code interpreters and LaTeX renderers.
- External tools can also make unimodal models multimodal.
- These extensions boost performance with fewer resources than finetuning.

Write Action Tools

- Write actions allow agents to modify their environment.
- Examples include sending emails, executing SQL updates, or initiating transactions.
- These actions enable automation of complete workflows.
- However, write actions increase risks of security breaches or harmful outcomes.
- Proper safeguards are critical to responsible deployment.

Planning in Agents

- Planning determines how agents sequence tool use to complete tasks.
- Agents must break complex tasks into subtasks and reason step by step.
- Planning involves both short-term reasoning and long-term strategy.
- Common methods include chain-of-thought, self-critique, and structured workflows.
- Strong planners help reduce compound errors across multiple steps.

Evaluating Agents

- Agents are more complex to evaluate than static models.
- Evaluations must consider success rates across multi-step tasks.
- Key risks include compounding errors, increased costs, and higher latency.
- Safety evaluation is critical for agents with write actions.
- Human-in-the-loop monitoring may be necessary in early deployments.

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References

References

- Huyen, Chip. AI engineering: Building applications with foundation models. O'Reilly Media, Inc., 2024.
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